# Qwen3.8-Flash-Next: which quant should YOU run?

10 files · 3 labs · 20 effort arms · one measured RTX 5070 Ti (16 GB).

Ryzen 9800X3D · 64 GB RAM · Windows + WSL2 | Oct 7–12, 2026 | Engine column: Strata 0.1.40.3 / AgentionAI rc1 / rc2

Blue = ISTA-DASLab · Orange = UkisAI · Green = AgentionAI | PR #1850 · revised 2026-10-12

## IN A HURRY? The study in six lines

- **Best measured stock all-rounder:** Q-Q2_0 (ISTA-DASLab) xhigh is #1 on the corrected eligible scoreboard (72). This is a seven-axis choice aid, not a universal intelligence score.
- **Pick the specialist when one axis matters.** Stock agent reliability: S-Q2_0 (UkisAI) low (78.3 floor). Fast coding: S-IQ3_XXS (UkisAI) low (11m56s). Those lower-composite picks do not contradict the all-rounder.
- **Coding ceilings are close; costs are not.** Highest complete reported score 96.3%: Q-IQ3_S (ISTA-DASLab) xhigh, Gyro-S (AgentionAI) low. Swift IQ3_S low reported the same percentage with only 163 scored tasks and is qualified, not tied on equal coverage.
- **Tool support is a serving-stack fact.** Gyro-M / rc2 joined the same pinned tool evaluator at 86.0 low and 87.8 xhigh. Gyro-S / rc1 remains n/a for tools. That is not an isolated model-only comparison.
- **More thinking has mixed returns.** Gyro-M xhigh solved one additional coding task after repair at 4.8x wall and 4.9x tokens. Its Hard Mode mean rose, but its floor fell from 78.3 to 71.7.
- **Fit and validity are part of the recommendation.** Gyro-M is experimental on this 16 GB card: 205 MiB minimum free VRAM, 78 C peak, invalid-slice status. No daily-driver endorsement. Partial scoring and test failures remain disclosed.

## THE FOUR QUESTIONS — straight answers

### Best overall on the measured stock rig?

**Q-Q2_0 (ISTA-DASLab) xhigh · #1 overall**. Same winner in the final board and first Pick Your File card.

Equal-weight composite; specialists can be better at the job you actually need.

### Fastest complete coding loop?

**S-IQ3_XXS (UkisAI) low** · #9 overall. 11m56s at 95.1%.

This is elapsed coding-run time, not the fastest decoder.

### Best stock agent reliability?

**S-Q2_0 (UkisAI) low** · #13 overall. Floor 78.3.

Gyro-M low ties the observed floor, but is a safety-capped experimental fork; no stock-composite rank.

### Best debugging / knowledge sample?

**S-IQ3_S (UkisAI)**: hidden debugging 20/25, algebra sample 85%.

Low coding arm is partial/unranked; xhigh is #12. A specialist, not the all-round crown.

PR #1850

## THE CAST — ten files, one base model, three lineages

All files share Qwen3.8-Flash-Next ancestry (Qwen / Alibaba). Q- labels: ISTA-DASLab quants; S- labels: UkisAI Swift derivatives; Gyro: AgentionAI APR. File sizes are decimal GB, not VRAM requirements. Results are integrated in the same tables; the engine and validity columns preserve comparison boundaries.

**T01 · Model provenance and engine cohorts**

| File / lab | Quant | File GB | Engine | Status | Download |
|---|---|---|---|---|---|
| **Q-IQ3_XXS** ISTA-DASLab | IQ3_XXS | 75.8 | 0.1.40.3 | reference | HF repo |
| **Q-Q2_0** ISTA-DASLab | Q2_0 | 66.4 | 0.1.40.3 | reference | HF repo |
| **Q-IQ3_S** ISTA-DASLab | IQ3_S | 83.6 | 0.1.40.3 | reference | HF repo |
| **Q-IQ2_XS** ISTA-DASLab | IQ2_XS | 67.2 | 0.1.40.3 | reference | HF repo |
| **S-IQ3_XXS** UkisAI | IQ3_XXS | 76.0 | 0.1.40.3 | reference | HF repo |
| **S-IQ2_XS** UkisAI | IQ2_XS | 68.2 | 0.1.40.3 | reference | HF repo |
| **S-IQ3_S** UkisAI | IQ3_S | 83.7 | 0.1.40.3 | reference | HF repo |
| **S-Q2_0** UkisAI | Q2_0 | 66.6 | 0.1.40.3 | reference | HF repo |
| **Gyro-S** AgentionAI | TQ1_0 | 58.5 | rc1 | historical fork | HF repo |
| **Gyro-M** AgentionAI | TQ2_0 | 92.0 | rc2 | experimental | HF repo |

T01 · 10 rows · end of table

The eight reference files use stock Strata 0.1.40.3; Gyro-S uses historical AgentionAI rc1, Gyro-M rc2. Shared evaluator and prompts do not remove the engine/configuration variable. Fork rows are inline observations, not ranked stock-engine controls.

## THE SCORECARD — all models in one view

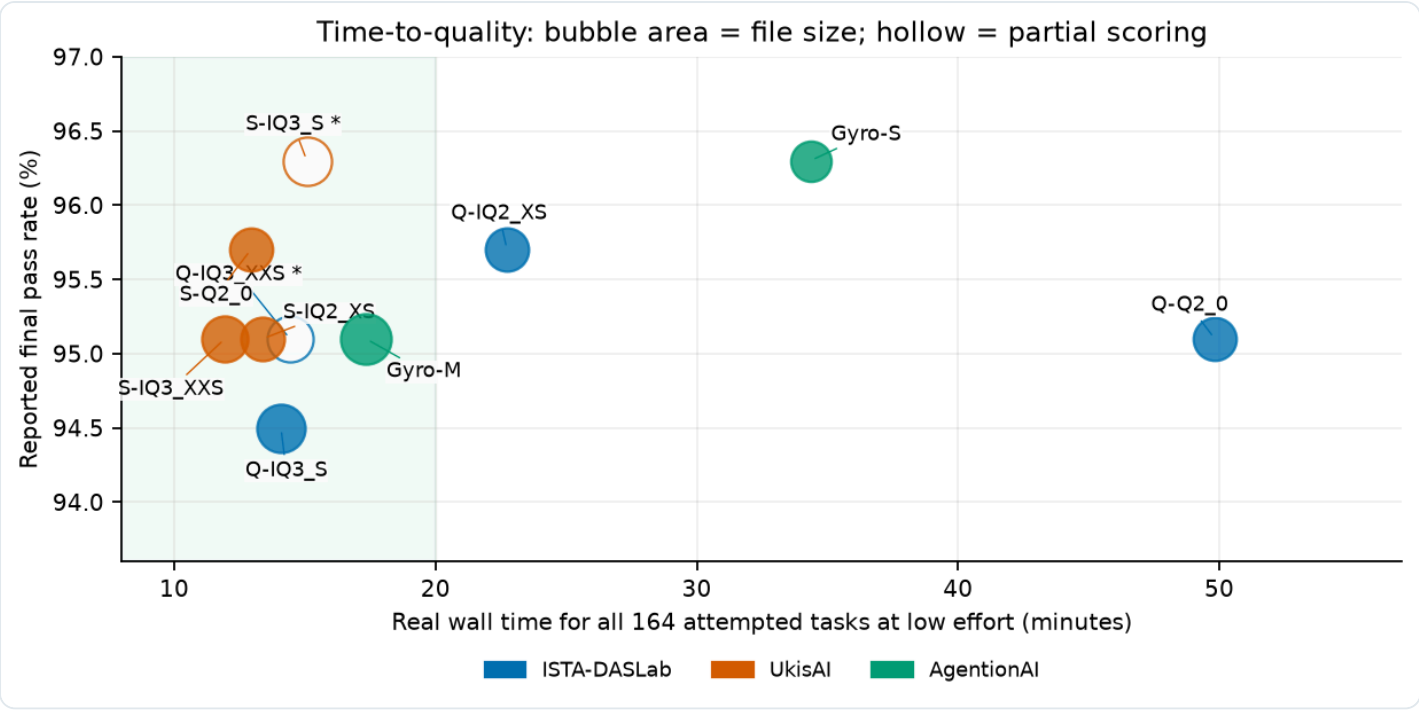
Tools: low-effort mean of five runs and empirical all-five floor. Coding: final percentage plus the actual solved/scored count. Asterisk = partial scoring. Knowledge/debugging are limited once-per-file tests. Stock rows sorted by floor; fork observations follow, with engine labels and no stock-composite rank.

**T02 · Integrated headline scorecard**

| File / lab | Engine | Tool mean | Pass^5 | HE+ low | HE+ xhigh | Math | Algebra | Debug |
|---|---|---|---|---|---|---|---|---|
| **S-Q2_0** UkisAI | 0.1.40.3 | 85.0 +/- 1.0 | 78.3 1 | 95.7 (157/164) | 93.3 (153/164) | 90 | 72 | 18/25 |
| **Q-IQ3_XXS** ISTA-DASLab | 0.1.40.3 | 87.0 +/- 2.0 | 76.1 2 | 95.1 (155/163)* | 95.7 (157/164) | 99 | 79 | 16/25 |
| **S-IQ3_S** UkisAI | 0.1.40.3 | 88.2 +/- 1.5 | 76.1 2 | 96.3 (157/163)* | 94.5 (155/164) | 98 | 85 | 20/25 |
| **S-IQ3_XXS** UkisAI | 0.1.40.3 | 84.6 +/- 0.5 | 75.0 3 | 95.1 (156/164) | 95.1 (156/164) | 96 | 80 | 19/25 |
| **S-IQ2_XS** UkisAI | 0.1.40.3 | 87.2 +/- 1.3 | 75.0 3 | 95.1 (156/164) | 94.5 (155/164) | 97 | 83 | 18/25 |
| **Q-IQ3_S** ISTA-DASLab | 0.1.40.3 | 86.8 +/- 0.8 | 73.9 | 94.5 (155/164) | 96.3 (158/164) | 97 | 82 | 19/25 |
| **Q-IQ2_XS** ISTA-DASLab | 0.1.40.3 | 86.2 +/- 0.8 | 73.9 | 95.7 (157/164) | 94.5 (155/164) | 97 | 83 | 19/25 |
| **Q-Q2_0** ISTA-DASLab | 0.1.40.3 | 84.8 +/- 0.4 | 70.7 | 95.1 (156/164) | 95.1 (156/164) | 97 | 78 | 19/25 |
| **Gyro-M** AgentionAI | rc2 | 86.0 +/- 1.4 | 78.3 | 95.1 (156/164) | 95.7 (157/164) | 96 | 79 | 18/25 |
| **Gyro-S** AgentionAI | rc1 | n/a | n/a | 96.3 (158/164) | 94.4 (153/162)* | 92 | 76 | 16/25 |

T02 · 10 rows · end of table

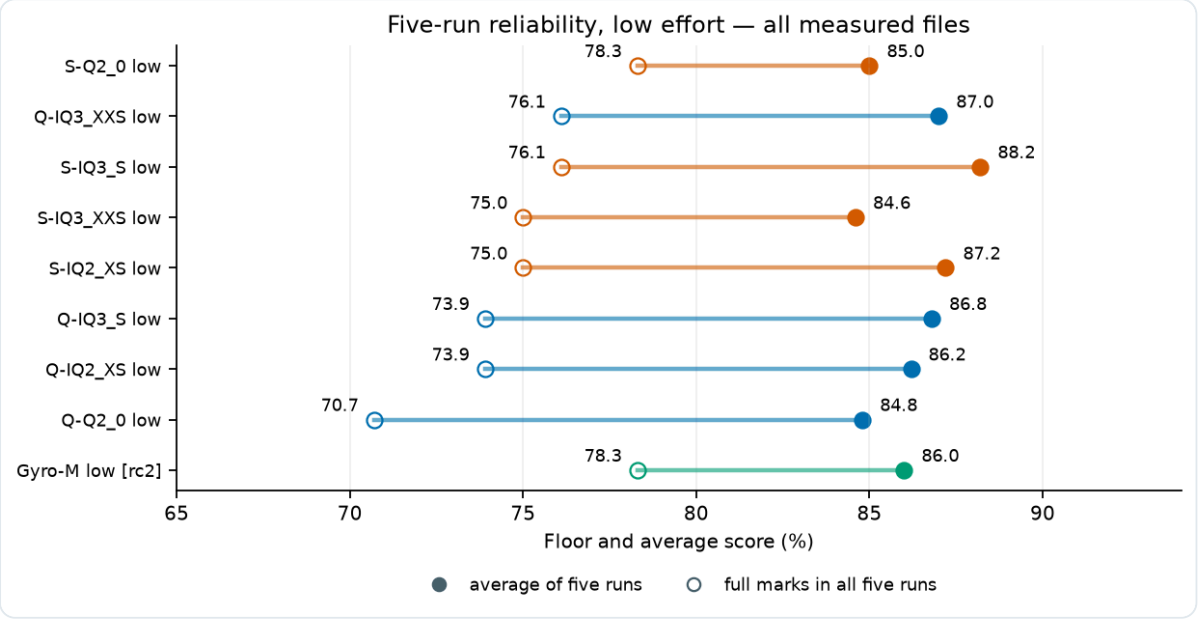
Medals show the top three distinct stock-reference score levels, ties share. Fork rows do not receive stock medals. n/a is an intentional unavailable measurement, not zero.




Low-effort observed time-to-quality. Hollow points have partial scoring; green points use separate fork engines. No model-only causal speedup is claimed.

## RELIABILITY & SAFETY — averages are not guarantees

Tool-eval-bench: 92 realistic scenarios x five identical-seed runs at each effort. Score 0 / 1 / 2; 2 = full marks. Pass@5: full marks at least once, the empirical capability ceiling. Pass^5: full marks in ALL five, the reliability floor. Partial credit every run never passes the floor. These summaries are not deployment guarantees.



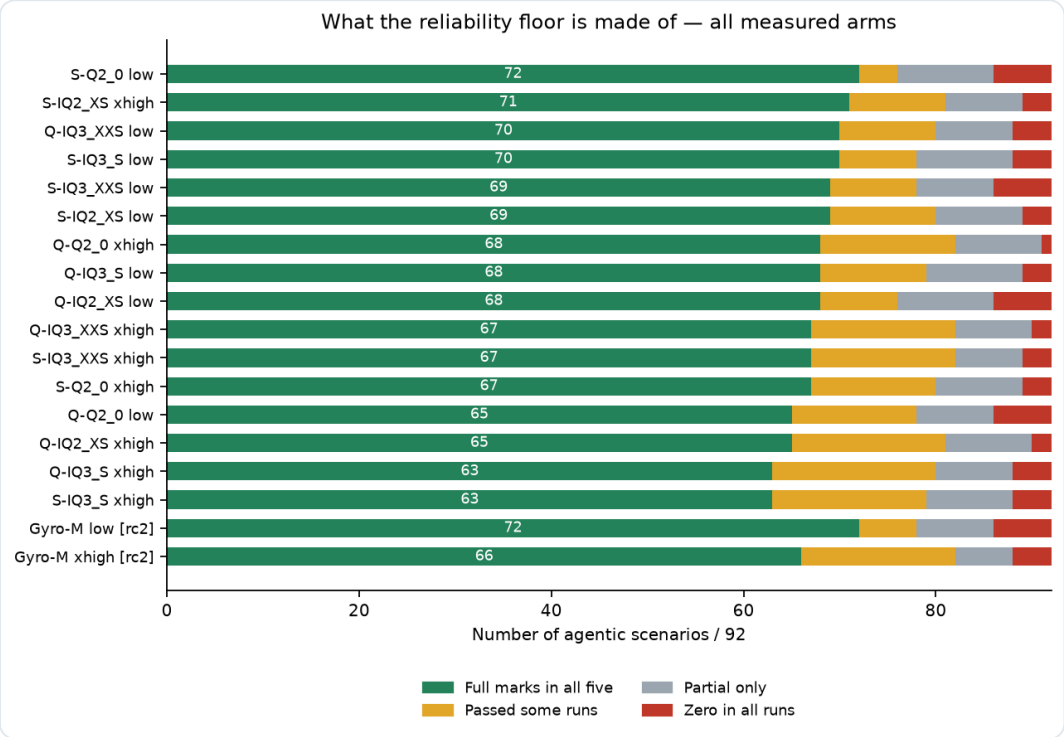

All measured low-effort files in the same reliability view. Engine labels preserve the stock/fork distinction. The line joins mean and floor; the table gap is Pass@5 minus Pass^5.

**T03 · Integrated five-trial tool reliability**

| Arm / lab | Mean +/- SD | 95% CI | Pass@5 | Pass^5 floor | Ceiling-floor gap | Last-trial rating | Warnings | Engine |
|---|---|---|---|---|---|---|---|---|
| **S-Q2_0** UkisAI low | 85.0 +/- 1.0 | 84.2–85.8 | 82.6 | 78.3 1 | 4.3 | 4/5 Good | 5 | 0.1.40.3 |
| **S-IQ2_XS** UkisAI xhigh | 88.2 +/- 1.5 3 | 87.0–89.4 | 88.0 | 77.2 2 | 10.8 | 5/5 Excellent | 2 | 0.1.40.3 |
| **Q-IQ3_XXS** ISTA-DASLab low | 87.0 +/- 2.0 | 85.4–88.6 | 87.0 | 76.1 3 | 10.9 | 3/5 safety-capped | 7 | 0.1.40.3 |
| **S-IQ3_S** UkisAI low | 88.2 +/- 1.5 3 | 87.0–89.2 | 84.8 | 76.1 3 | 8.7 | 4/5 Good | 4 | 0.1.40.3 |
| **S-IQ3_XXS** UkisAI low | 84.6 +/- 0.5 | 84.2–85.0 | 84.8 | 75.0 | 9.8 | 4/5 Good | 4 | 0.1.40.3 |
| **S-IQ2_XS** UkisAI low | 87.2 +/- 1.3 | 86.2–88.2 | 87.0 | 75.0 | 12.0 | 3/5 safety-capped | 4 | 0.1.40.3 |
| **Q-Q2_0** ISTA-DASLab xhigh | 89.4 +/- 0.5 1 | 89.0–89.8 | 89.1 | 73.9 | 15.2 | 5/5 Excellent | 2 | 0.1.40.3 |
| **Q-IQ3_S** ISTA-DASLab low | 86.8 +/- 0.8 | 86.2–87.4 | 85.9 | 73.9 | 12.0 | 4/5 Good | 3 | 0.1.40.3 |
| **Q-IQ2_XS** ISTA-DASLab low | 86.2 +/- 0.8 | 85.6–86.8 | 82.6 | 73.9 | 8.7 | 3/5 safety-capped | 5 | 0.1.40.3 |
| **Q-IQ3_XXS** ISTA-DASLab xhigh | 88.6 +/- 1.1 2 | 87.8–89.4 | 89.1 | 72.8 | 16.3 | 3/5 safety-capped | 3 | 0.1.40.3 |
| **S-IQ3_XXS** UkisAI xhigh | 86.6 +/- 1.5 | 85.4–87.8 | 89.1 | 72.8 | 16.3 | 3/5 safety-capped | 4 | 0.1.40.3 |
| **S-Q2_0** UkisAI xhigh | 86.8 +/- 2.2 | 85.2–88.4 | 87.0 | 72.8 | 14.2 | 4/5 Good | 2 | 0.1.40.3 |
| **Q-Q2_0** ISTA-DASLab low | 84.8 +/- 0.4 | 84.4–85.0 | 84.8 | 70.7 | 14.1 | 3/5 safety-capped | 7 | 0.1.40.3 |
| **Q-IQ2_XS** ISTA-DASLab xhigh | 88.6 +/- 1.8 2 | 87.2–90.0 | 88.0 | 70.7 | 17.3 | 4/5 Good | 1 | 0.1.40.3 |
| **Q-IQ3_S** ISTA-DASLab xhigh | 86.8 +/- 1.3 | 85.6–87.8 | 87.0 | 68.5 | 18.5 | 3/5 safety-capped | 4 | 0.1.40.3 |
| **S-IQ3_S** UkisAI xhigh | 85.6 +/- 0.5 | 85.2–86.0 | 85.9 | 68.5 | 17.4 | 3/5 safety-capped | 5 | 0.1.40.3 |
| **Gyro-M** AgentionAI low | 86.0 +/- 1.4 | 84.8–87.2 | 84.8 | 78.3 | 6.5 | 3/5 safety-capped | 6 | rc2 |
| **Gyro-M** AgentionAI xhigh | 87.8 +/- 1.1 | 86.8–88.6 | 89.1 | 71.7 | 17.4 | 4/5 Good | 2 | rc2 |

T03 · 18 rows · end of table

8/16 stock arms are capped on the stored last-trial rating. Gyro-M adds one capped and one Good rating. Not capped is NOT the same as no warnings or production safety. Warning lists/rating describe the final primary trial, not a five-run union.



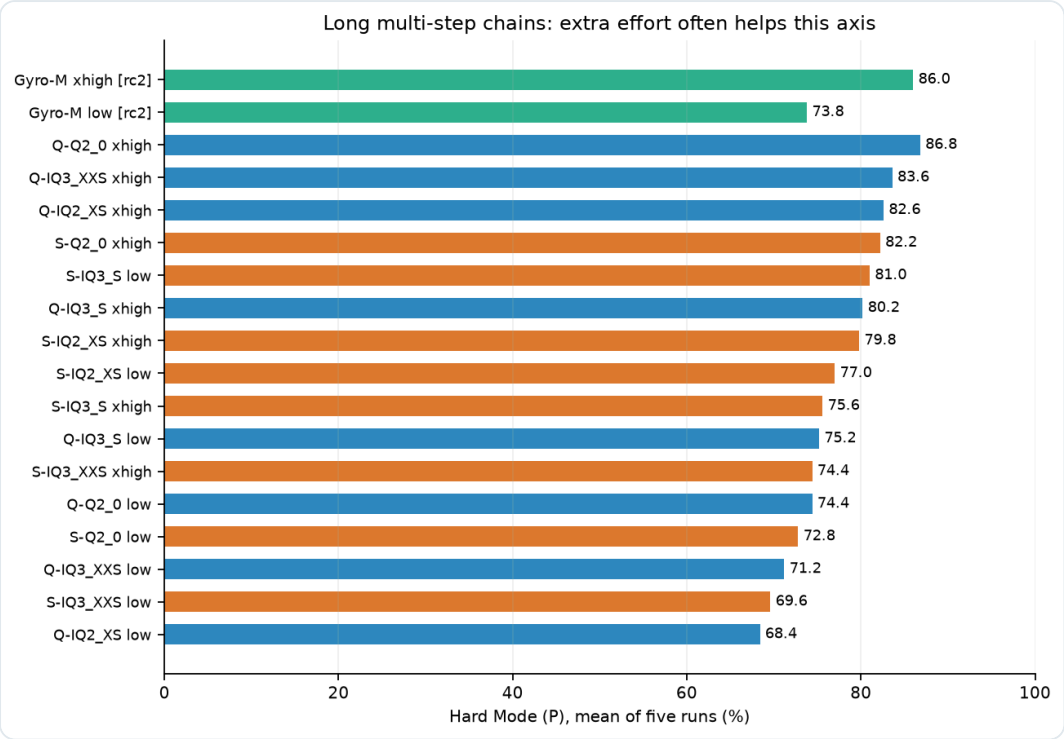

The all-five floor is the green count / 92. All tool-capable arms appear; fork rows are labeled rc2. A higher mean can coexist with more run-to-run flakes.

> Gyro-M illustrates the trade-off: always-full scenarios 72 low / 66 xhigh; flaky 6 / 16. Its mean confidence intervals overlap, so the observed mean gain is modest. No causal mechanism is inferred from the aggregate alone.

Tool-eval-bench

# AGENTIC TOOL USE — hard chains reward extra thought

Hard Mode (P) stresses long chains and recovery. Every category figure uses the five-run mean, not the last-trial score. Effort comparisons stay within a measured file/build. The engine column prevents the fork data from masquerading as a stock quant A/B.




All measured arms in one Hard Mode chart. Stock ranking and rc2 observational rows are distinguished by their labels.

**T04 · Multi-step tool-chain results**

| Arm / lab | Hard Mode mean | SD across five | Turn budget | Engine |
|---|---|---|---|---|
| **Q-Q2_0** ISTA-DASLab xhigh | 86.8 (1) | 4.8 | 8 | 0.1.40.3 |
| **Q-IQ3_XXS** ISTA-DASLab xhigh | 83.6 (2) | 2.6 | 8 | 0.1.40.3 |
| **Q-IQ2_XS** ISTA-DASLab xhigh | 82.6 (3) | 4.9 | 8 | 0.1.40.3 |
| **S-Q2_0** UkisAI xhigh | 82.2 | 4.3 | 8 | 0.1.40.3 |
| **S-IQ3_S** UkisAI low | 81.0 | 3.1 | 8 | 0.1.40.3 |
| **Q-IQ3_S** ISTA-DASLab xhigh | 80.2 | 3.9 | 8 | 0.1.40.3 |
| **S-IQ2_XS** UkisAI xhigh | 79.8 | 3.0 | 8 | 0.1.40.3 |
| **S-IQ2_XS** UkisAI low | 77.0 | 5.1 | 8 | 0.1.40.3 |
| **S-IQ3_S** UkisAI xhigh | 75.6 | 3.0 | 8 | 0.1.40.3 |
| **Q-IQ3_S** ISTA-DASLab low | 75.2 | 1.1 | 8 | 0.1.40.3 |
| **Q-Q2_0** ISTA-DASLab low | 74.4 | 2.2 | 8 | 0.1.40.3 |
| **S-IQ3_XXS** UkisAI xhigh | 74.4 | 3.3 | 8 | 0.1.40.3 |
| **S-Q2_0** UkisAI low | 72.8 | 3.3 | 8 | 0.1.40.3 |
| **Q-IQ3_XXS** ISTA-DASLab low | 71.2 | 7.2 | 8 | 0.1.40.3 |
| **S-IQ3_XXS** UkisAI low | 69.6 | 3.6 | 8 | 0.1.40.3 |
| **Q-IQ2_XS** ISTA-DASLab low | 68.4 | 2.3 | 8 | 0.1.40.3 |
| **Gyro-M** AgentionAI xhigh | 86.0 | 4.7 | 8 | rc2 |
| **Gyro-M** AgentionAI low | 73.8 | 4.4 | 8 | rc2 |

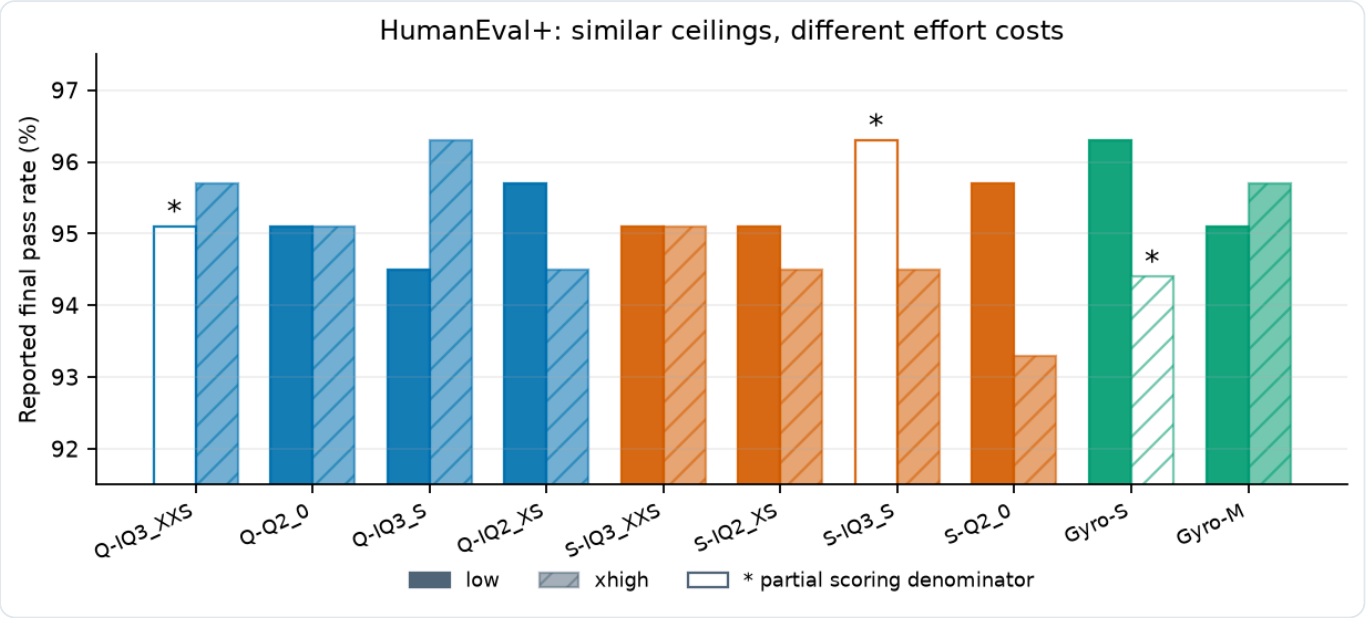
T04 · 18 rows · end of table

- Gyro-M Hard Mode rises from 73.8% to 86.0%, while its all-five floor falls. Optimizing only one of those axes is not a reliable workload decision.
- Warnings include premature notifications, missed dependency ordering and claims that an invoice is paid while funds remain held. They are scripted failure traps, not permissions.

- rc2 live forced tool calls passed streaming and non-streaming with Gyro-M; Gyro-S rc1 remains unavailable for tool-eval. This does not establish Gyro-S behavior on rc2.

## CODE — every denominator visible

164 function-writing tasks via BenchKit / EvalPlus, greedy generation, one verifier-guided repair. First = one-shot score; final includes the replacement. Scored timeout/length/blank outputs remain failures. Excluded harness errors can reduce the denominator; those rows retain observed percentages but are qualified and excluded from full-coverage/composite claims.




All ten files; hollow starred bars use 163 or 162 scored tasks. Other ordinary failures remain in the 164-task denominator.

**T05 · Integrated coding results — full and qualified coverage**

| Arm / lab | First % | Final % | Solved / scored | Repairs | Wall | tok/s | Blank finals | Infra excluded | Engine |
|---|---|---|---|---|---|---|---|---|---|
| **Q-IQ3_S** ISTA-DASLab xhigh | 93.3 | 96.3 1 | 158/164 | 5/11 | 67m19s | 86.7 | 2 | 0 | 0.1.40.3 |
| **Q-IQ3_XXS** ISTA-DASLab xhigh | 92.7 | 95.7 2 | 157/164 | 5/11 | 76m17s | 102.6 | 3 | 0 | 0.1.40.3 |
| **Q-IQ2_XS** ISTA-DASLab low | 90.9 | 95.7 2 | 157/164 | 8/14 | 22m44s | 124.5 | 2 | 0 | 0.1.40.3 |
| **S-Q2_0** UkisAI low | 90.2 | 95.7 2 | 157/164 | 9/16 | 12m56s 2 | 144.8 | 0 | 0 | 0.1.40.3 |
| **Q-Q2_0** ISTA-DASLab low | 92.7 | 95.1 3 | 156/164 | 4/10 | 49m50s | 123.9 | 2 | 0 | 0.1.40.3 |
| **Q-Q2_0** ISTA-DASLab xhigh | 88.4 | 95.1 3 | 156/164 | 11/17 | 62m18s | 124.0 | 3 | 0 | 0.1.40.3 |
| **S-IQ3_XXS** UkisAI low | 92.7 | 95.1 3 | 156/164 | 4/12 | 11m56s 1 | 123.7 | 0 | 0 | 0.1.40.3 |
| **S-IQ3_XXS** UkisAI xhigh | 92.1 | 95.1 3 | 156/164 | 5/10 | 47m11s | 104.5 | 5 | 0 | 0.1.40.3 |
| **S-IQ2_XS** UkisAI low | 89.6 | 95.1 3 | 156/164 | 9/16 | 13m23s 3 | 148.8 | 1 | 0 | 0.1.40.3 |
| **Q-IQ3_S** ISTA-DASLab low | 92.7 | 94.5 | 155/164 | 3/12 | 14m05s | 100.4 | 0 | 0 | 0.1.40.3 |
| **Q-IQ2_XS** ISTA-DASLab xhigh | 89.6 | 94.5 | 155/164 | 8/16 | 75m40s | 122.0 | 6 | 0 | 0.1.40.3 |
| **S-IQ2_XS** UkisAI xhigh | 91.5 | 94.5 | 155/164 | 5/11 | 63m04s | 124.0 | 5 | 0 | 0.1.40.3 |
| **S-IQ3_S** UkisAI xhigh | 92.7 | 94.5 | 155/164 | 3/11 | 43m32s | 89.5 | 3 | 0 | 0.1.40.3 |
| **S-Q2_0** UkisAI xhigh | 89.6 | 93.3 | 153/164 | 6/14 | 56m20s | 121.8 | 6 | 0 | 0.1.40.3 |
| **S-IQ3_S** UkisAI low | 94.5 | 96.3* | 157/163 | 3/9 | 15m05s | 101.5 | 0 | 1 | 0.1.40.3 |
| **Q-IQ3_XXS** ISTA-DASLab low | 92.6 | 95.1* | 155/163 | 4/13 | 14m26s | 121.5 | 0 | 1 | 0.1.40.3 |
| **Gyro-S** AgentionAI low | 92.1 | 96.3 | 158/164 | 7/13 | 34m21s | 71.7 | 2 | 0 | rc1 |
| **Gyro-M** AgentionAI xhigh | 92.1 | 95.7 | 157/164 | 6/13 | 82m51s | 77.6 | 2 | 0 | rc2 |
| **Gyro-M** AgentionAI low | 93.3 | 95.1 | 156/164 | 3/11 | 17m20s | 88.9 | 0 | 0 | rc2 |
| **Gyro-S** AgentionAI xhigh | 90.1 | 94.4* | 153/162 | 7/16 | 329m15s | 63.1 | 3 | 2 | rc1 |

T05 · 20 rows · end of table

Retrospective coverage correction: Q-IQ3_XXS low 155/163; Swift IQ3_S low 157/163; Gyro-S xhigh 153/162 after two HTTP 502 harness errors. Their observed percentages are preserved, not silently rescored. Gyro-M xhigh blank finals on HumanEval/39 and HumanEval/99 ended on length and remain scored failures; 157/164 is not a clean-response certificate.

- Gyro-M xhigh costs 4.8x low wall and 4.9x tokens for one additional final pass. Hard Mode benefits do not imply a coding-efficiency benefit.
- Gyro-M low vs historical Gyro-S low has 1.2 points more first-attempt score, 1.2 points less final score. Engines/configurations differ; this is not isolated quant improvement.

EvalPlus BenchKit

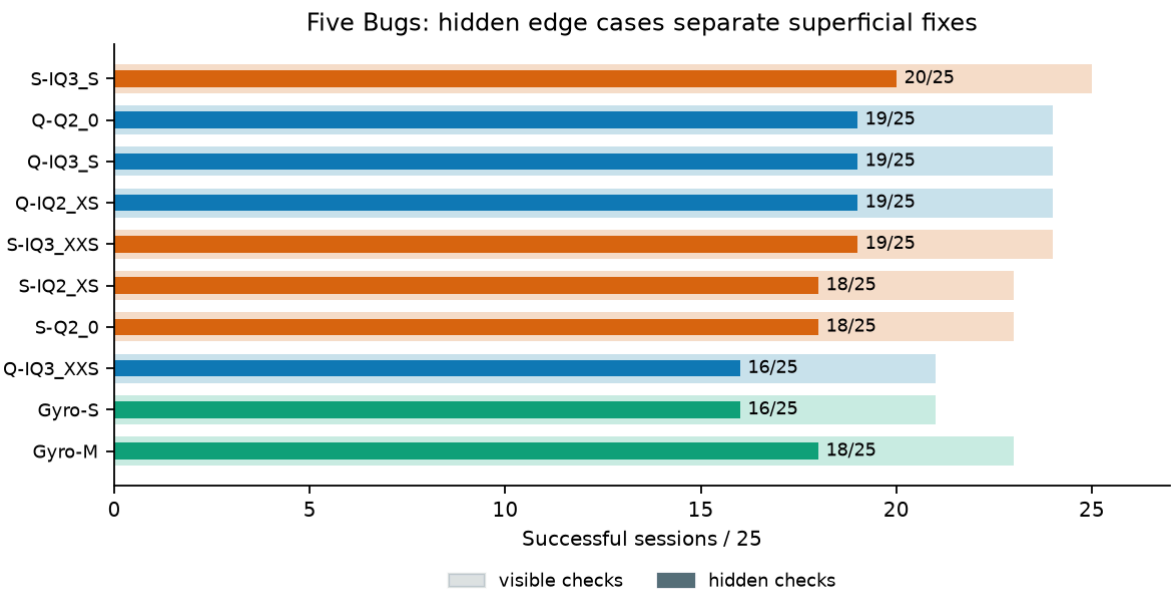
## KNOWLEDGE & DEBUGGING — the test we actually ran

GSM8K: 100 seeded-shuffled questions, 8-shot. MMLU: first 100, 5-shot, entirely abstract algebra — NOT a 57-subject exam. IFEval: first 50 prompts / 76 constraints. Five Bugs: five buggy Python files x five sessions, visible plus local hidden checks. All are limited samples; do not turn them into a whole-model intelligence ranking.

**T06 · Integrated sampled knowledge, debugging and long-context recall**

| File / lab | GSM8K /100 | Algebra /100 | IFEval prompt / instruction % | Bugs visible | Bugs hidden | Needle % @ tokens | Engine |
|---|---|---|---|---|---|---|---|
| **S-IQ3_S** UkisAI | 98 | 85 | 84.0 / 89.5 | 25/25 | 20/25 1 | 100 @ 260,096 | 0.1.40.3 |
| **Q-Q2_0** ISTA-DASLab | 97 | 78 | 84.0 / 89.5 | 24/25 | 19/25 2 | 100 @ 260,096 | 0.1.40.3 |
| **Q-IQ3_S** ISTA-DASLab | 97 | 82 | 86.0 / 90.8 | 24/25 | 19/25 2 | 100 @ 260,096 | 0.1.40.3 |
| **Q-IQ2_XS** ISTA-DASLab | 97 | 83 | 84.0 / 89.5 | 24/25 | 19/25 2 | 100 @ 260,096 | 0.1.40.3 |
| **S-IQ3_XXS** UkisAI | 96 | 80 | 80.0 / 86.8 | 24/25 | 19/25 2 | 100 @ 260,096 | 0.1.40.3 |
| **S-IQ2_XS** UkisAI | 97 | 83 | 82.0 / 88.2 | 23/25 | 18/25 3 | 100 @ 260,096 | 0.1.40.3 |
| **S-Q2_0** UkisAI | 90 | 72 | 80.0 / 86.8 | 23/25 | 18/25 3 | 100 @ 260,096 | 0.1.40.3 |
| **Q-IQ3_XXS** ISTA-DASLab | 99 | 79 | 80.0 / 85.5 | 21/25 | 16/25 | 100 @ 260,096 | 0.1.40.3 |
| **Gyro-M** AgentionAI | 96 | 79 | 78.0 / 82.9 | 23/25 | 18/25 | 100 @ 129,024 | rc2 |
| **Gyro-S** AgentionAI | 92 | 76 | 74.0 / 82.9 | 21/25 | 16/25 | 100 @ 129,024 | rc1 |

T06 · 10 rows · end of table




Every model in the same visible/hidden debugging plot. Fork results retain engine-qualified observational status.

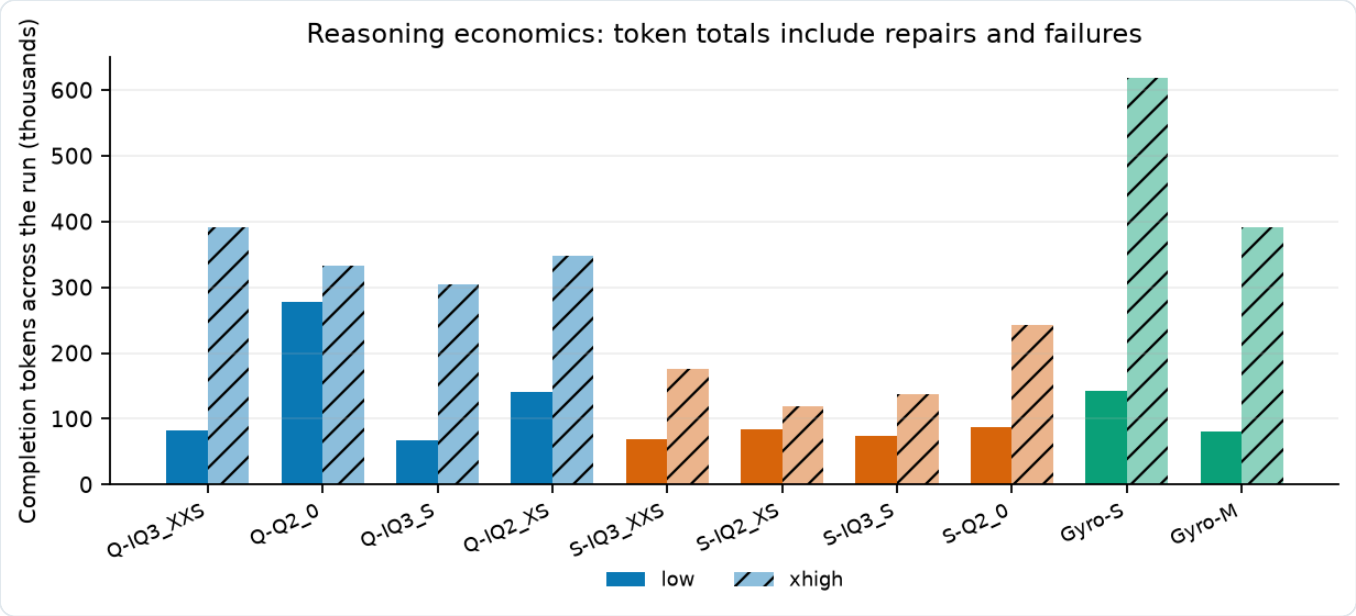
S-IQ3_S (UkisAI) remains the debugging specialist at 20/25 hidden sessions and leads this algebra sample at 85%. Its low coding arm is partial and cannot claim the stock all-round crown.

Needle: 25 cells (five lengths x five insertion depths). Effective retrieved lengths are 260,096 for stock and 129,024 for both fork rows. Gyro-M configured context 131,072; no 260K claim for it.

Five Bugs upstream | TEB knowledge / needle plugins

# REASONING ECONOMICS — what you wait for

Total output tokens include reasoning, repairs and failures. Wall includes request/harness overhead. Failure time share and passing-task cost come from task-level response times, not wall subtraction. Different engine/protocol rows remain descriptive.




Completion-token totals for all attempted tasks, all files, both efforts. Partial-scoring arms retain their coverage qualifier.

**T07 · Integrated time-to-result economics**

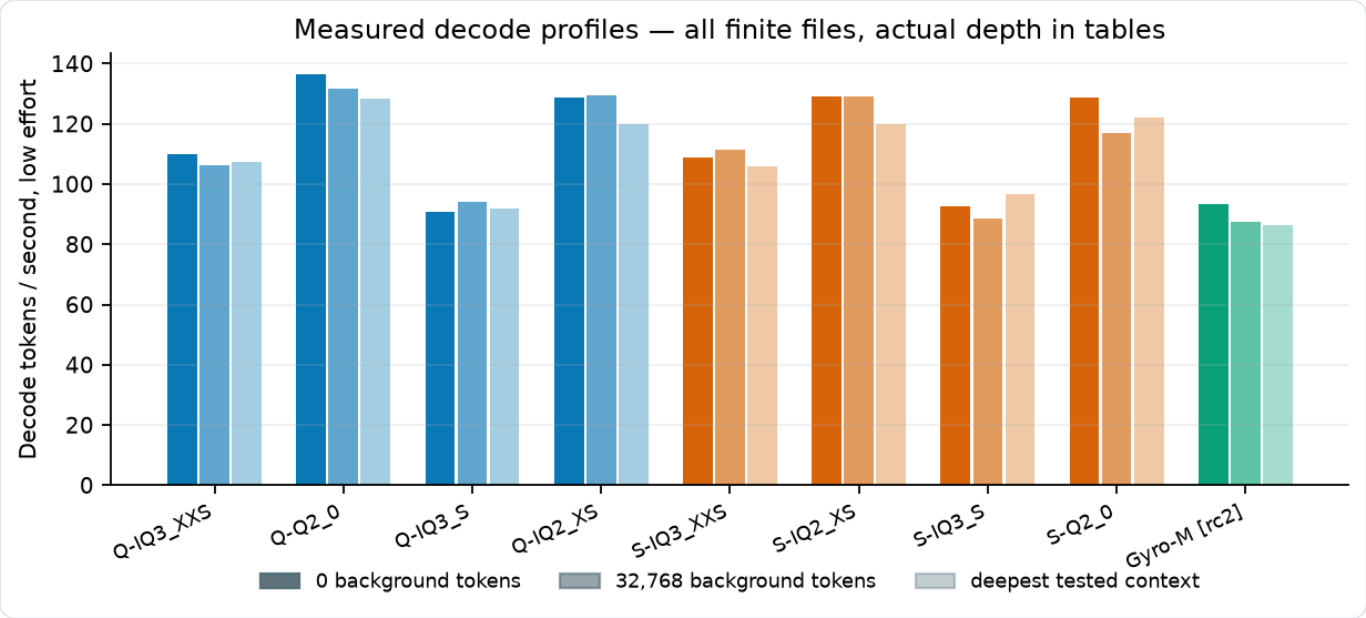
| Arm / lab | Output tokens | Passing tokens | Passing seconds | Failed response time % | Wall | Engine |
|---|---|---|---|---|---|---|
| **S-IQ3_XXS** UkisAI low | 68,616 | 49,192 | 486 | 25.1% | 11m56s | 0.1.40.3 |
| **S-Q2_0** UkisAI low | 87,634 | 64,502 | 542 | 23.4% | 12m56s | 0.1.40.3 |
| **S-IQ2_XS** UkisAI low | 83,746 | 56,476 | 468 | 36.3% | 13m23s | 0.1.40.3 |
| **Q-IQ3_S** ISTA-DASLab low | 66,967 | 49,239 | 594 | 23.9% | 14m05s | 0.1.40.3 |
| **Q-IQ3_XXS** ISTA-DASLab low | 82,843 | 59,863 | 602 | 24.7% | 14m26s | 0.1.40.3 |
| **S-IQ3_S** UkisAI low | 74,417 | 49,341 | 598 | 29.0% | 15m05s | 0.1.40.3 |
| **Q-IQ2_XS** ISTA-DASLab low | 140,583 | 97,239 | 883 | 32.1% | 22m44s | 0.1.40.3 |
| **S-IQ3_S** UkisAI xhigh | 137,839 | 81,261 | 1,017 | 60.2% | 43m32s | 0.1.40.3 |
| **S-IQ3_XXS** UkisAI xhigh | 175,860 | 133,095 | 1,363 | 50.7% | 47m11s | 0.1.40.3 |
| **Q-Q2_0** ISTA-DASLab low | 277,084 | 174,148 | 1,498 | 48.8% | 49m50s | 0.1.40.3 |
| **S-Q2_0** UkisAI xhigh | 243,330 | 151,062 | 1,329 | 60.0% | 56m20s | 0.1.40.3 |
| **Q-Q2_0** ISTA-DASLab xhigh | 332,296 | 238,341 | 2,020 | 45.1% | 62m18s | 0.1.40.3 |
| **S-IQ2_XS** UkisAI xhigh | 118,588 | 105,555 | 2,472 | 33.6% | 63m04s | 0.1.40.3 |
| **Q-IQ3_S** ISTA-DASLab xhigh | 304,962 | 200,835 | 2,461 | 38.1% | 67m19s | 0.1.40.3 |
| **Q-IQ2_XS** ISTA-DASLab xhigh | 347,704 | 257,698 | 2,214 | 50.5% | 75m40s | 0.1.40.3 |
| **Q-IQ3_XXS** ISTA-DASLab xhigh | 390,930 | 230,123 | 2,328 | 48.5% | 76m17s | 0.1.40.3 |
| **Gyro-M** AgentionAI low | 80,523 | 60,016 | 752 | 23.2% | 17m20s | rc2 |
| **Gyro-S** AgentionAI low | 143,288 | 69,471 | 1,316 | 34.1% | 34m21s | rc1 |
| **Gyro-M** AgentionAI xhigh | 391,467 | 180,234 | 2,334 | 52.5% | 82m51s | rc2 |
| **Gyro-S** AgentionAI xhigh | 618,982 | 375,211 | 6,148 | 60.1% | 329m15s | rc1 |

T07 · 20 rows · end of table

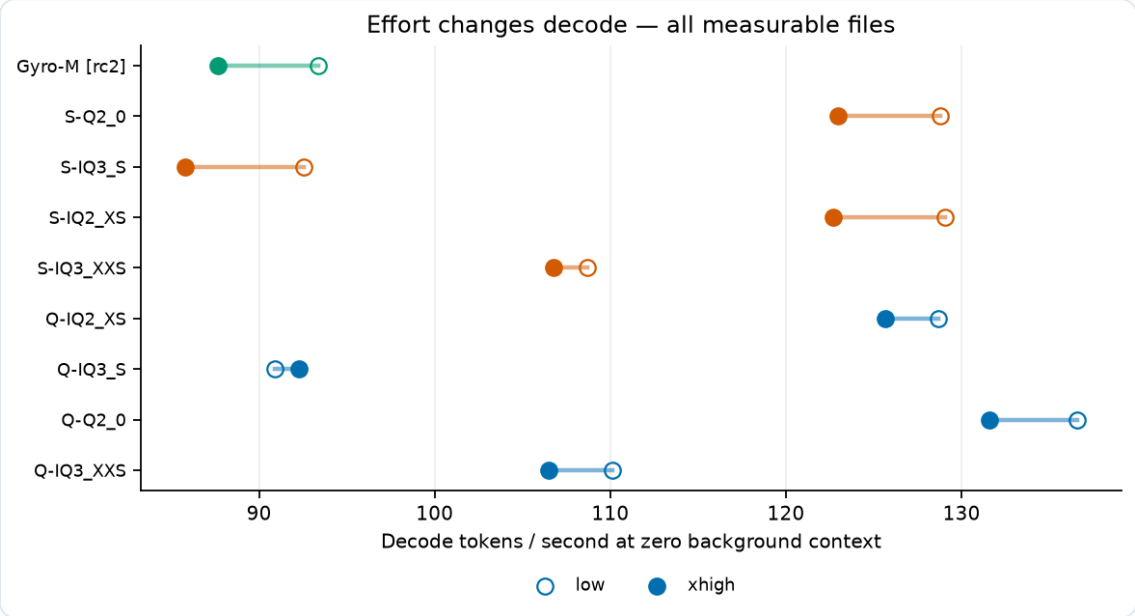
Pick effort for the task: more deliberation can improve long tool chains while making code loops slower. The overall ranking averages capability axes, not latency or safety; the specialist cards below say exactly which axis overrides that ranking.

## SPEED — real decode; corrected first-token timing

llama-benchy 0.4.0: prompt 2,048, generation 256, concurrency 1, three repeats/depth/effort. Raw pp_throughput in the old PDF was inflated by early SSE role chunks and is retired. Here prompt-to-first-token rate = (prompt + background context) / e2e TTFT, including network/queue/reasoning. It is NOT an engine-kernel prefill benchmark. Largest depths differ: stock 131,072; rc2 122,880.




All finite low-effort decode profiles. Lightest bar = deepest measured point; row tables state the exact token depths. Gyro-S malformed profiles remain n/a.




Decode at zero background context for both efforts, all measurable files. A fork-engine observation is not a same-engine model A/B.

**T08 · Integrated low-effort performance profiles**

| File / lab | Engine | Context tokens: 0 / mid / deep | Decode tok/s: 0 / mid / deep | First-token rate: 0 / mid / deep | Deep TTFT s |
|---|---|---|---|---|---|
| Q-Q2_0 ISTA-DASLab | 0.1.40.3 | 0 / 32,768 / 131,072 | 136.6 / 131.8 / 128.4 | 2,309 / 3,522 / 3,667 | 36.30 |
| S-IQ2_XS UkisAI | 0.1.40.3 | 0 / 32,768 / 131,072 | 129.1 / 129.1 / 119.9 | 1,970 / 3,283 / 3,334 | 39.93 |
| S-Q2_0 UkisAI | 0.1.40.3 | 0 / 32,768 / 131,072 | 128.8 / 116.9 / 121.9 | 1,892 / 3,229 / 3,329 | 39.99 |
| Q-IQ2_XS ISTA-DASLab | 0.1.40.3 | 0 / 32,768 / 131,072 | 128.7 / 129.4 / 120.0 | 2,011 / 3,195 / 3,280 | 40.59 |
| Q-IQ3_XXS ISTA-DASLab | 0.1.40.3 | 0 / 32,768 / 131,072 | 110.1 / 106.2 / 107.4 | 2,174 / 3,284 / 3,421 | 38.91 |
| S-IQ3_XXS UkisAI | 0.1.40.3 | 0 / 32,768 / 131,072 | 108.7 / 111.3 / 106.0 | 2,025 / 3,329 / 3,412 | 39.01 |
| S-IQ3_S UkisAI | 0.1.40.3 | 0 / 32,768 / 131,072 | 92.6 / 88.5 / 96.5 | 1,594 / 3,206 / 3,374 | 39.45 |
| Q-IQ3_S ISTA-DASLab | 0.1.40.3 | 0 / 32,768 / 131,072 | 90.9 / 94.0 / 91.7 | 1,641 / 3,164 / 3,041 | 43.77 |
| Gyro-M AgentionAI | rc2 | 0 / 32,768 / 122,880 | 93.4 / 87.5 / 86.5 | 1,239 / 2,144 / 2,821 | 44.28 |
| Gyro-S AgentionAI | rc1 | n/a | n/a | n/a | n/a |

T08 · 10 rows · end of table

**T09 · Integrated xhigh-effort performance profiles**

| File / lab | Engine | Context tokens: 0 / mid / deep | Decode tok/s: 0 / mid / deep | First-token rate: 0 / mid / deep | Deep TTFT s |
|---|---|---|---|---|---|
| Q-Q2_0 ISTA-DASLab | 0.1.40.3 | 0 / 32,768 / 131,072 | 131.6 / 130.5 / 117.2 | 2,676 / 3,836 / 3,663 | 36.35 |
| Q-IQ2_XS ISTA-DASLab | 0.1.40.3 | 0 / 32,768 / 131,072 | 125.7 / 118.9 / 117.5 | 2,135 / 3,341 / 3,339 | 39.87 |
| S-Q2_0 UkisAI | 0.1.40.3 | 0 / 32,768 / 131,072 | 123.0 / 121.9 / 128.9 | 2,219 / 3,167 / 3,332 | 39.96 |
| S-IQ2_XS UkisAI | 0.1.40.3 | 0 / 32,768 / 131,072 | 122.7 / 123.9 / 113.8 | 2,209 / 3,286 / 3,334 | 39.93 |
| S-IQ3_XXS UkisAI | 0.1.40.3 | 0 / 32,768 / 131,072 | 106.8 / 101.1 / 99.4 | 2,420 / 3,339 / 3,449 | 38.59 |
| Q-IQ3_XXS ISTA-DASLab | 0.1.40.3 | 0 / 32,768 / 131,072 | 106.5 / 101.7 / 95.3 | 2,406 / 3,361 / 3,409 | 39.05 |
| Q-IQ3_S ISTA-DASLab | 0.1.40.3 | 0 / 32,768 / 131,072 | 92.2 / 91.2 / 86.6 | 1,945 / 3,216 / 3,382 | 39.36 |
| S-IQ3_S UkisAI | 0.1.40.3 | 0 / 32,768 / 131,072 | 85.8 / 85.5 / 85.8 | 1,933 / 3,269 / 3,388 | 39.29 |
| Gyro-M AgentionAI | rc2 | 0 / 32,768 / 122,880 | 87.7 / 80.5 / 80.7 | 1,754 / 2,690 / 2,778 | 44.96 |
| Gyro-S AgentionAI | rc1 | n/a | n/a | n/a | n/a |

T09 · 10 rows · end of table

> Gyro-S synthetic streaming produced million-token/s decode artifacts and null depth cells. Those profiles are excluded, not recycled as a valid speed baseline. Every printed finite profile has three stored measurements per depth; both efforts appear in the same format.

# THE FINAL SCOREBOARD — one message, with explicit scope

OVERALL averages seven min-max-normalized axes within the 14 eligible stock-engine arms: coding final, agent floor, Hard Mode mean, algebra sample, math sample, hidden debugging, zero-context decode. Full scoring only. Forks and partial arms appear inline but are unranked. This crowns the best measured STOCK ALL-ROUNDER, not every workload winner; safety and latency are not averaged into that score.

Decision board: stock-engine composite only; forks and partial arms unranked

| | Rank / OVERALL | Coding HE+ | Agents Pass^5 | Hard Mode | MMLU sample | GSM8K sample | Debug /25 | Decode tok/s | Rating gate |
|---|---|---|---|---|---|---|---|---|---|
| Q-Q2_0 xhigh | 1. 72 | 95.1 | 73.9 | 86.8 | 78 | 97 | 19 | 132 | not capped |
| S-IQ2_XS xhigh | 2. 68 | 94.5 | 77.2 | 79.8 | 83 | 97 | 18 | 123 | not capped |
| S-IQ2_XS low | 3. 67 | 95.1 | 75.0 | 77.0 | 83 | 97 | 18 | 129 | capped |
| Q-IQ2_XS low | 4. 65 | 95.7 | 73.9 | 68.4 | 83 | 97 | 19 | 129 | capped |
| Q-IQ2_XS xhigh | 5. 65 | 94.5 | 70.7 | 82.6 | 83 | 97 | 19 | 126 | not capped |
| Q-Q2_0 low | 6. 59 | 95.1 | 70.7 | 74.4 | 78 | 97 | 19 | 137 | capped |
| Q-IQ3_S xhigh | 7. 58 | 96.3 | 68.5 | 80.2 | 82 | 97 | 19 | 92 | capped |
| Q-IQ3_XXS xhigh | 8. 57 | 95.7 | 72.8 | 83.6 | 79 | 99 | 16 | 106 | capped |
| S-IQ3_XXS low | 9. 54 | 95.1 | 75.0 | 69.6 | 80 | 96 | 19 | 109 | not capped |
| S-IQ3_XXS xhigh | 10. 54 | 95.1 | 72.8 | 74.4 | 80 | 96 | 19 | 107 | capped |
| Q-IQ3_S low | 11. 53 | 94.5 | 73.9 | 75.2 | 82 | 97 | 19 | 91 | not capped |
| S-IQ3_S xhigh | 12. 53 | 94.5 | 68.5 | 75.6 | 85 | 98 | 20 | 86 | capped |
| S-Q2_0 low | 13. 48 | 95.7 | 78.3 | 72.8 | 72 | 90 | 18 | 129 | not capped |
| S-Q2_0 xhigh | 14. 35 | 93.3 | 72.8 | 82.2 | 72 | 90 | 18 | 123 | not capped |
| Q-IQ3_XXS low | unranked | 95.1* | 76.1 | 71.2 | 79 | 99 | 16 | 110 | capped |
| S-IQ3_S low | unranked | 96.3* | 76.1 | 81.0 | 85 | 98 | 20 | 93 | not capped |
| Gyro-S low | unranked | 96.3 | n/a | n/a | 76 | 92 | 16 | n/a | n/a |
| Gyro-S xhigh | unranked | 94.4* | n/a | n/a | 76 | 92 | 16 | n/a | n/a |
| Gyro-M low | unranked | 95.1 | 78.3 | 73.8 | 79 | 96 | 18 | 93 | capped |
| Gyro-M xhigh | unranked | 95.7 | 71.7 | 86.0 | 79 | 96 | 18 | 88 | not capped |

Same ranks and scope used by every recommendation below. Asterisk = partial coding denominator. Not capped is a final-trial rating status, not a safety guarantee.

**All-rounder:** Q-Q2_0 (ISTA-DASLab) xhigh · #1 overall (72). **Specialists:** S-Q2_0 (UkisAI) low · #13 owns the stock floor; S-IQ3_XXS (UkisAI) low · #9 owns the fastest complete coding wall. Pick Your File translates THIS board; it does not introduce a second overall winner.

**T10 · One recommendation registry — all-rounder versus specialists**

| Decision | File / effort | OVERALL rank | Why this agrees with the board |
|---|---|---|---|
| Best stock all-rounder | **Q-Q2_0** ISTA-DASLab xhigh | #1 | Balance the seven measured capability axes |
| Reliability specialist | **S-Q2_0** UkisAI low | #13 | Prioritize the empirical tool floor |
| Coding-latency specialist | **S-IQ3_XXS** UkisAI low | #9 | Prioritize complete-run wall time, outside OVERALL |
| Coding-ceiling specialist | **Q-IQ3_S** ISTA-DASLab xhigh | #7 | Prioritize the complete coding score |
| Knowledge / debugging specialist | **S-IQ3_S** UkisAI xhigh | #12 | Prioritize independent sampled tests; low is partial/unranked |
| Experimental observation only | **Gyro-M** AgentionAI low | unranked | Different engine and narrow 16 GB margin; no production pick |

T10 · 6 rows · end of table

## PICK YOUR FILE — the scoreboard translated into jobs

**One decision rule:** use the #1 stock all-rounder if you balance the seven measured axes. Override that composite only when your workload has a named priority — reliability, coding latency, coding ceiling, or debugging. Lower-ranked specialists are deliberate exceptions, not alternative overall crowns. Unranked experimental/partial rows are not eligible production recommendations.

### Balanced workload · best stock all-rounder

**Q-Q2_0 (ISTA-DASLab) xhigh · #1 overall**

Score 72 across the seven eligible stock axes. Also leads stock tool mean at 89.4.

Composite is equal-weighted, on one rig and limited knowledge samples; not a universal quality/safety claim.

### Reliability-first agent · specialist override

**S-Q2_0 (UkisAI) low · #13 overall**

Stock floor 78.3 is highest. Choose this lower-ranked arm when consistent full-mark tool behavior outweighs the other axes.

Still has scripted safety warnings. Gyro-M low ties the observed floor but is unranked/experimental, not an equivalent production pick.

### Fast coding loop · specialist override

**S-IQ3_XXS (UkisAI) low · #9 overall**

Complete coding wall 11m56s at 95.1%.

You are prioritizing latency, which is outside the composite; this is not a new all-round winner.

### Highest stock coding ceiling · specialist override

**Q-IQ3_S (ISTA-DASLab) xhigh · #7 overall**

Complete reported coding 96.3% (158/164).

Scored blank/timeout failures remain. Gyro-S low ties this reported ceiling on rc1, but is not a tool-capable stock control.

### Knowledge / debugging · specialist override

**S-IQ3_S (UkisAI)**

Hidden checks 20/25, algebra sample 85%. Low unranked (partial coding); xhigh #12 overall.

The partial low coding result prevents an overall crown. Specialist evidence is from separate debugging/knowledge samples; choose it only for that priority.

### Experimental fork · no production recommendation

**Gyro-M (AgentionAI) low · unranked rc2**

HE+ 95.1%, observed floor 78.3, live tools. The board displays its results but does not authorize a stock rank.

Minimum 205 MiB free VRAM, 131,072 context, invalid-slice status. Not a supported 16 GB daily-driver pick. Gyro-S rc1 has no valid tools score.

## FOR MAINTAINERS — measured findings and validity

- **Tool plumbing is build-specific:** rc1 forced calls were ignored with Gyro-S. rc2 live forced calls passed with Gyro-M on streaming and non-streaming. The historical finding stays scoped to rc1; Gyro-S on rc2 was not tested.
- **Experimental fork fit:** 74 resource samples, minimum 205 MiB free, peak 78 C. cuda_error=0, engine_restart=0, no_progress=0, stall_report=0, verify_timeout=0. This is not long-term stability proof.
- **Campaign status:** `completed_with_invalid_slices`, finished 2026-10-12T17:27:48+02:00. CTest 49 pass / 2 skip / 1 fail. ple_parity lacks its required Q2_0 PLE shard/capture fixtures; separate ple_reader_selftest passed.
- **Audit bookkeeping:** both HumanEval+ task audits ran; a dictionary merge overwrote intended audit-step names, making done.json mark them absent. Xhigh validity logic also missed blank finals. Raw status/scores are preserved; no silently upgraded clean pass.
- **Monitoring:** corrected busy-vs-down logic after the first aborted attempt. Perf-low/needle static-counter warnings were not accompanied by engine-event counters. Standalone effort probe is excluded from scores; possible timing interference is disclosed.
- **Operations:** production Swift IQ3_S restoration passed health/chat/tool/vision and re-enabled guards/updater. Report revisions did not alter services.

# METHOD & CORRECTIONS — reproducible without hiding caveats

The PDF, GitHub README, scoreboard and recommendation registry are generated from the same extracted store. New data is integrated throughout, not isolated in a model-specific chapter. Source records and hashes are packaged; the portable compiler rebuilds from this folder.

- **Coding:** BenchKit direct, 164 attempted tasks, greedy temperature 0, presence penalty 1.0, one repair. Gyro-M timeout 900 seconds, no injected client token cap. Serving protocol and engine changes qualify historical cross-cohort comparisons.
- **Tools:** 2.7.1.dev16+gf7c34a130, 92 scenarios x five trials per effort; seed 42, temperature 0.0, timeout 300 s, turn cap 8. Separate Swift v2.8.0 / 97-scenario data is NOT mixed in.
- **Corrections:** historical partial coding denominators qualified; category scores changed to five-run means; inflated raw prefill removed; ceiling-minus-floor gap labeled correctly; peak temperature corrected from 64 C summary to measured 78 C. These corrections, not an unmeasured model promotion, change the earlier recommendation framing.
- **Not tested:** vision quality, multi-day agents, extra seeds, multi-GPU, other GPU sizes, full knowledge datasets, Gyro-M beyond 131K or Gyro-S on rc2. No automatic supported-memory or production-stability conclusion.

**T11 · Version-pinned provenance**

| Provenance item | Pinned value |
|---|---|
| Stock engine digest | 34CDE150B21148E6 |
| Fork rc2 run | full-20261012-rerun-1230 |
| Fork engine commit | `9ff02c9fa9c86a88088370925b3c9131cc5d6c26` |
| Fork binary SHA256 | `08d70e30b9176dee5273ba74bd67c4d4c435bef92d449323168d7a8a3497585f` |
| Gyro-M model revision | `2cd4f24bcb0fa053149a4e4ec15506ecc64cafab` |
| Gyro-M model SHA256 | `7aa36d15131d255179ab6c87fa9c9f81a0b541fc9e81e959d4c3e29eb20feb5f` |
| Fork context / KV | 131,072 / int8 |
| Rebuild | scripts/build_revision.py; data/ALL-METRICS.json; claims.json |

T11 · 8 rows · end of table

PR #1850 · Strata · AgentionAI fork · Gyro source · Tool-eval-bench · EvalPlus · Five Bugs

Revised 2026-10-12 · source-qualified observations · all recommendations use the same registry and ranks. Numeric audit, whole-table pagination proof, source manifest and portable rebuild accompany the report.